MAHMOUD AHMED ABDELAWAL

AI & MACHINE LEARNING ENGINEER • TECHNICAL PROJECT DOSSIER

Location: Sohag, Egypt | **DOB:** March 15, 2004 | **Military Status:** Exempted
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1. EXECUTIVE SUMMARY & DISTINCTIONS

AI & Machine Learning Engineer and Computer Science graduate with hands-on experience developing end-to-end AI solutions across Computer Vision, Deep Learning, NLP, and Generative AI. Proficient in Python, TensorFlow, PyTorch, and FastAPI, with practical experience building multimodal AI systems, integrating machine learning models, and developing RAG-based applications. Key projects include a five-model AI-powered disaster management system (Grade A+) and an award-winning clinical decision support platform.

■ **1st Place Champion — AI Hackathon:** Awarded First Place nationwide among engineering teams across Egypt with *DermaAssist AI* (WHO-grounded RAG + Multimodal Dermatology Decision Support), sponsored by the Ministry of Communications & IT, Orange Egypt, and Instant.

■■ **Ministerial Recognition for Excellence:** Formally recognized by the **Minister of Communications and Information Technology (MCIT) of Egypt** as a top-performing student nationwide in the Digital Egypt Pioneers Initiative (DEPI).

■ **Professional Experience:** AI Data Trainer (Freelance / Contract) on **Crowdgen Platform** (2025–2026), evaluating and fine-tuning generative AI models, data annotation, and logical reasoning benchmarks.

2. PROJECT 01 — S-REPORT AI: UNIVERSAL URBAN INTELLIGENCE

Category: Graduation Project (Grade A+) | Repository: github.com/MahmoudMahrous24/S-Report-AI-Vision

S-Report AI is a state-of-the-art hybrid multimodal computer vision and spatial intelligence platform (Graduation Project awarded **Grade A+**) engineered for real-time disaster detection, damage severity evaluation, and automated emergency resource dispatch.

Technical Architecture & Core Innovations (5-Model Ensemble):

- **Hybrid Zero-Shot Hazard Classification:** Combines OpenAI's CLIP (Contrastive Language-Image Pre-training) zero-shot visual-linguistic embeddings with a fine-tuned ConvNeXt-Tiny convolutional backbone for high-accuracy, sub-150ms classification of critical emergency situations (fire, building collapse, major flooding, vehicle accidents).
- **Spatial Scene Understanding (BLIP):** Integrates BLIP (Bootstrapping Language-Image Pre-training) to automatically synthesize rich natural language scene descriptions and situational context for first responders.
- **Monocular 3D Depth Estimation (MiDaS):** Incorporates MiDaS relative depth estimation models to construct 3D depth maps, allowing spatial hazard proximity calculations and structural collapse risk assessment.
- **Real-Time Object & Hazard Localization (YOLOv8):** Deploys YOLOv8 instance segmentation to detect, count, and outline trapped civilians, emergency vehicles, and infrastructure hazards in real time.

- **Weighted Decision & Dispatch Engine:** Engineered a mathematical multi-criteria triage algorithm that aggregates hazard severity, civilian density, and proximity into an automated priority dispatch score.
- **Production-Ready Infrastructure:** Built with an asynchronous FastAPI RESTful backend, SQLite database persistence, and Docker containerization ready for cloud deployment.

Core Stack: Python, PyTorch, CLIP, ConvNeXt-Tiny, YOLOv8, BLIP, MiDaS, OpenCV, FastAPI, Docker, SQLite, NumPy

3. PROJECT 02 — DERMAASSIST AI: CLINICAL DECISION SUPPORT

Distinction: ■ 1st Place Champion — Nationwide AI Hackathon | Repository: github.com/MahmoudMahrous24/AI-Hackathon-Sohag

DermaAssist AI is an award-winning bilingual (Arabic/English) clinical decision support platform designed to bridge the specialist healthcare deficit in dermatology, strictly grounded in official World Health Organization (WHO) clinical guidelines.

Technical Architecture & Core Innovations:

- **WHO-Grounded RAG Pipeline:** Built an evidence-based Retrieval-Augmented Generation pipeline using ChromaDB vector database and Nomic Embeddings indexed over certified WHO dermatological clinical manuals, eliminating hallucination risks.
- **Advanced LLM Orchestration:** Integrated Google Gemini Flash via structured prompt engineering to generate accurate, bilingual diagnostic assistance, clinical rationales, and emergency warnings.
- **Multimodal Skin Lesion Analysis:** Implemented deep learning vision models capable of analyzing high-resolution dermoscopic images and live video streams across 3 input modalities (image, video, voice).
- **Dual-Portal Medical Platform:** Developed separate authenticated interfaces for patients and certified dermatologists, featuring automated prescription generation, electronic health records (EHR), and triage management.

Core Stack: React 19, TypeScript, FastAPI, Gemini Flash, ChromaDB, Nomic Embeddings, SentenceTransformers, SQLite, WebRTC

4. PROJECT 03 — ARTIFICIAL VINCENT VAN GOGH: GENERATIVE ART AI

Category: DEPI Graduation Project | Repository: github.com/DEPI-Project/Artificial-Vincent-Van-Gogh

Artificial Vincent Van Gogh is a generative deep learning pipeline engineered to synthesize authentic artwork capturing the post-impressionist brushstroke dynamics, color palettes, and texture nuances of Vincent Van Gogh.

Technical Architecture & Core Innovations:

- **DCGANs Architecture:** Trained Deep Convolutional Generative Adversarial Networks from scratch on a curated high-resolution dataset of Van Gogh paintings, utilizing strided convolutions, batch normalization, and LeakyReLU activations.
- **Neural Style Transfer (NST):** Formulated optimization algorithms leveraging VGG-19 feature representations and Gram Matrix spatial correlations to transfer post-impressionist artistic style onto arbitrary natural photographs.

Core Stack: Python, PyTorch, Torchvision, DCGANs, VGG-19, OpenCV, Matplotlib, NumPy, CUDA

5. PROJECT 04 — END-TO-END DATA ANALYSIS & STATISTICAL MODELING

Category: Data Science Portfolio | Repository: github.com/MahmoudMahrous24/Data-Analysis

A comprehensive collection of data analysis pipelines focused on exploratory data analysis (EDA), anomaly detection, statistical hypothesis testing, data cleaning, and actionable business insight extraction across real-world datasets.

Technical Architecture & Methodologies:

- **Exploratory Data Analysis (EDA):** Systematic profiling of dataset distributions, multivariate correlations, skewness metrics, and feature interactions.
- **Data Cleaning & Preprocessing:** Robust handling of missing data, outlier identification via IQR and Z-score methods, categorical encoding, and feature scaling.
- **Statistical Testing & Insights:** Conducting hypothesis tests to extract statistically sound conclusions and guide predictive modeling.
- **Visual Storytelling:** Publication-quality charts using Matplotlib, Seaborn, and Pandas.

Core Stack: Python, Pandas, NumPy, Scikit-Learn, SciPy, Matplotlib, Seaborn, Jupyter Notebooks

6. COMPREHENSIVE SKILLS & CERTIFICATIONS MATRIX

Domain	Key Technologies & Frameworks
AI & Deep Learning	Machine Learning, Deep Learning, Neural Networks, CNNs, GANs, Transfer Learning, Model Training, PyTorch, TensorFlow, Scikit-Learn
Computer Vision	YOLOv8, Instance Segmentation, Image Classification, OpenCV, MiDaS Depth Estimation, CLIP, BLIP, ConvNeXt
NLP & Generative AI	Text Preprocessing, Tokenization, Embeddings, Sequence Modeling, RAG Pipelines, ChromaDB Vector DB, Gemini API
Languages & Soft Skills	Python (Expert), Java, C++, JavaScript, TypeScript, SQL | Arabic: Native, English: Professional Working Proficiency | Problem Solving, Teamwork
Backend & DevOps	FastAPI, React, SQLite, Git, GitHub, Microsoft Azure, Docker, VS Code
Certifications & Courses	Data Analysis Diploma (Creativa Hub, Oct-Nov 2026) • Deep Learning, CV & NLP (Hisham Asem, 2025) • Microsoft ML Engineer (DEPI, 2024) • English & Soft Skills (DEPI, 2024) • Huawei HCIA-AI V3.5 (2024)